MACHINE LEARNING LAB EXERCISES

Topic 1: Matplotlib, Statistics and Probability

A. Matplotlib – Data Visualization

Q1. Scatter Plot: Write a Python program to create a scatter plot showing the relationship between hours studied and marks obtained by 10 students. Add a suitable title, x-axis label, y-axis label, and grid.

Q2. Line Plot: Write a Python program to plot the daily temperatures of a city over seven days using a line graph. Display markers at each data point and add a legend.

Q3. Bar Graph: A college records the number of students enrolled in five departments: CSE (120), ECE (90), Mechanical (75), Civil (60), and Electrical (80). Write a Python program to display the data using a bar graph.

Q4. Pie Chart: Write a Python program to display the percentage distribution of a student's monthly expenses: Food (30%), Rent (25%), Transport (15%), Education (20%), and Entertainment (10%). Display the percentage values on the chart.

Q5. Histogram: Generate 100 random marks between 0 and 100 using NumPy. Plot a histogram with 10 bins to show the distribution of marks. Add appropriate labels and a title.

Q6. Multiple Visualizations: Create a dataset containing the marks of 10 students in three subjects: Python, Mathematics, and Statistics. Use the following visualizations:

a) A bar graph to compare subject-wise marks.

b) A line plot to show the marks of each student across subjects.

c) A histogram to display the distribution of Python marks.

Q7. Real-World Visualization: Create a dataset containing monthly sales for a shop over 12 months. Visualize the data using a line plot and a bar graph. Identify the month with the highest sales.

B. Statistics and Probability

Q8. Measures of Central Tendency: Write a Python program using NumPy to calculate the mean, median, and mode of the following dataset:

[12, 15, 18, 20, 15, 22, 25, 15, 30, 28]

Q9. Measures of Dispersion: For the dataset [10, 20, 30, 40, 50], write a Python program to calculate the following measures:

a) Range

b) Variance

c) Standard deviation

d) Interquartile range (IQR)

Use NumPy wherever appropriate.

Q10. Comparing Two Datasets: The marks obtained by two students in five tests are given below:

Student A: [70, 75, 80, 85, 90]

Student B: [60, 80, 75, 90, 95]

Write a Python program to calculate the mean and standard deviation of both datasets. Determine which student has more consistent performance.

Q11. Normal Distribution: Use NumPy to generate 1,000 random values from a normal distribution with a mean of 50 and a standard deviation of 10. Plot a histogram and describe the shape of the distribution.

Q12. Uniform Distribution: Generate 1,000 random values from a uniform distribution between 0 and 1. Plot a histogram and compare its shape with the histogram obtained from a normal distribution.

Q13. Sampling: Create a NumPy array containing 100 integers between 1 and 500. Select a random sample of 10 elements without replacement. Calculate and compare the mean of the population and the sample.

Q14. Correlation: Create two arrays representing hours studied and marks obtained:

Hours: [1, 2, 3, 4, 5, 6]

Marks: [35, 40, 50, 55, 65, 75]

Write a Python program to calculate the Pearson correlation coefficient using NumPy. Interpret whether the relationship is positive, negative, or weak.

Q15. Correlation Visualization: Generate datasets representing positive, negative, and nearly zero correlation. Calculate the correlation coefficient for each pair and visualize the three relationships using scatter plots.

Q16. Integrated Statistics Exercise: Generate the marks of 50 students using a normal distribution with a mean of 65 and a standard deviation of 12. Calculate the mean, median, variance, and standard deviation. Plot a histogram and identify the number of students scoring above 80.

Topic 2: Machine Learning Fundamentals, NumPy and Pandas

A. Machine Learning vs Classical Programming

Q17. Classical Programming vs Machine Learning: Explain the difference between classical programming and machine learning with a suitable example. Write a Python program using conditional statements to determine whether a student passes or fails, and explain how this differs from a model trained on historical student data.

Q18. Ability to Predict: Consider a dataset containing house areas and house prices. Explain how a machine learning model can learn from this dataset and predict the price of a new house. Identify the features, target variable, training data, and prediction.

Q19. Rule-Based vs Data-Driven Prediction: Write a Python program that predicts whether a student passes based on a fixed rule of scoring at least 40 marks. Then explain how a machine learning classification model could learn a prediction rule from previous students' marks and outcomes.

Q20. Prediction Using Regression: Given the following dataset, write a Python program to train a simple linear regression model and predict the price of a house with an area of 1,500 square feet.

Area (sq. ft.) – Price (₹ lakh)

800 – 25

1000 – 30

1200 – 35

1400 – 40

1600 – 45

Use LinearRegression from sklearn.linear_model.

B. NumPy – Array Creation and Operations

Q21. Array Creation: Write a Python program using NumPy to perform the following operations:

a) Create a one-dimensional array containing integers from 1 to 10.

b) Create a 3 × 3 array containing integers from 1 to 9.

c) Create an array of five zeros and an array of five ones.

d) Display the data type and shape of each array.

Q22. Array Reshaping: Create a one-dimensional array containing numbers from 1 to 12. Reshape it into the following arrays:

a) A 3 × 4 matrix.

b) A 4 × 3 matrix.

c) A 2 × 2 × 3 array.

Display the shape of each resulting array.

Q23. Array Operations: Given the following arrays, write a Python program to perform element-wise addition, subtraction, multiplication, and division.

A = np.array([10, 20, 30, 40])

B = np.array([2, 4, 5, 8])

Q24. Indexing and Slicing: Create a NumPy array containing integers from 10 to 100 with a step size of 10. Extract the following:

a) The first three elements.

b) The last four elements.

c) Every second element.

d) The elements at indices 2, 4, and 6.

Q25. Matrix Operations: Create two 3 × 3 matrices using NumPy. Perform the following operations:

a) Matrix addition.

b) Matrix subtraction.

c) Element-wise multiplication.

d) Matrix multiplication.

e) Transpose of each matrix.

f) Row-wise and column-wise sums.

Q26. Conditional Array Operations: Given the following array of student marks:

marks = np.array([35, 78, 90, 42, 28, 65, 88, 50])

Write a Python program to perform the following operations:

a) Extract marks greater than or equal to 50.

b) Count students who scored below 40.

c) Replace all marks below 40 with 40.

d) Calculate the mean of the updated marks.

Q27. Broadcasting: Create a 3 × 3 NumPy matrix and add the one-dimensional array [10, 20, 30] to each row. Explain how NumPy broadcasting makes this operation possible.

C. Pandas – Data Handling and Analysis

For Questions 28–34, use the following dataset wherever appropriate.

```
import pandas as pd


data = {
    'Name': ['Amit', 'Riya', 'Rahul', 'Neha', 'Priya'],
    'Age': [20, 21, 19, 22, 20],
    'Marks': [85, 92, 67, 78, 88],
    'Department': ['CSE', 'ECE', 'CSE', 'IT', 'ECE']
}
```

df = pd.DataFrame(data)

Q28. DataFrame Creation and Summary: Write a Python program to perform the following operations:

a) Create the DataFrame.

b) Display the first three rows and last two rows.

c) Display the number of rows and columns.

d) Display column names and data types.

e) Generate a statistical summary using describe().

Q29. Indexing and Selection: Using the given DataFrame, write a Python program to perform the following operations:

a) Select the Name column.

b) Select the Name and Marks columns.

c) Display the first three rows using iloc.

d) Display rows with index labels 1 and 3 using loc.

Q30. Conditional Access: Write a Python program using the given DataFrame to perform the following operations:

a) Display students scoring more than 80 marks.

b) Display students whose age is at least 20.

c) Display CSE students scoring more than 70 marks.

d) Display students scoring between 70 and 90, inclusive.

Q31. Adding and Deleting Columns: Using the given DataFrame, write a Python program to perform the following operations:

a) Add a new column named Result, assigning Pass to students scoring at least 40 and Fail otherwise.

b) Add another column named Bonus, containing 5 extra marks for every student.

c) Delete the Bonus column.

d) Display the updated DataFrame.

Q32. Handling Missing Values: Create a DataFrame containing student names, marks, and ages, with missing values in some cells. Write a Python program to perform the following operations:

a) Identify missing values using isnull().

b) Count missing values in each column.

c) Replace missing marks with the mean marks.

d) Remove rows containing any remaining missing values.

e) Display the cleaned dataset.

Q33. Sorting and Filtering: Create a DataFrame containing the names and marks of 10 students. Write a Python program to perform the following operations:

a) Sort the DataFrame by marks in descending order.

b) Display the top three students.

c) Calculate the class average.

d) Filter and display students who scored above the class average.

Q34. Grouping and Aggregation: Create a DataFrame containing student names, departments, and marks. Group the data by department and calculate the following:

a) Average marks.

b) Maximum marks.

c) Minimum marks.

d) Number of students in each department.

Q35. Dataset Analysis: Download or create a CSV file containing student information with the columns Name, Age, Attendance, Assignment_Score, and Final_Marks. Write a Python program to perform the following operations:

a) Load the CSV file using read_csv().

b) Display dataset information and summary statistics.

c) Identify and handle missing values.

d) Find students with attendance above 75%.

e) Calculate the average final marks.

f) Investigate the relationship between assignment scores and final marks using correlation.

g) Visualize the relationship using a scatter plot.